

Multi-party Computation (MPC)

DSC 4110/6111: Modern Cryptography and AI/ML Security

Dr. Laltu Sardar

School of Data Science,
Indian Institute of Science Education and Research Thiruvananthapuram (IISER TVM)



October 5, 2026

Table of Content

- 1 MPC
- 2 Secret Sharing
- 3 Additive Secret Sharing
- 4 Threshold Secret Sharing

MPC

From Secure MLaaS to MPC

Our goal:

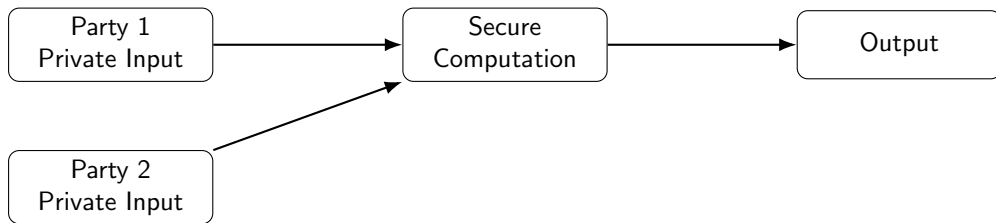
- User has a private query.
- Model owner has a private ML model.
- Cloud performs the computation.
- Neither party should reveal its private information.

Key Question

How can two or more parties compute together without revealing their private inputs?

Secure Multi-Party Computation (MPC)

MPC: The Basic Idea



- Parties contribute private inputs.
- They jointly evaluate a function.
- The required output is obtained.
- Private inputs remain protected.

A Simple Example

Suppose two hospitals want to compute:

Total number of patients satisfying a condition

Hospital A:

- Knows its own patient count.
- Does not want to reveal the count.

Hospital B:

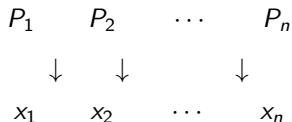
- Knows its own patient count.
- Does not want to reveal the count.

MPC

They obtain the total without revealing the individual counts.

Secure Multi-Party Computation

There are n Parties. Each party P_i holds a private input x_i .



The parties jointly compute:

$$y = f(x_1, x_2, \dots, x_n)$$

- The inputs $x_1, x_2, \dots, x_n$ remain private.
- Parties interact during the computation.
- The desired output y is obtained correctly.
- No party should learn the other parties' private inputs.

MPC Goal

Compute the function without revealing the inputs.

Important MPC Protocol Families

MPC is a broad area. Several important approaches exist:

- **Secret Sharing**

- Additive Secret Sharing
- Threshold Secret Sharing

- **Yao's Garbled Circuits**

- Two-party secure computation
- Boolean circuits

- **GMW Protocol**

- Multi-party Boolean computation

- **BGW Protocol**

- Secret sharing based
- Information-theoretic security

- **SPDZ**

- Practical MPC
- Supports malicious security

Secret Sharing

Secret Sharing: Problem Statement

Suppose a dealer has a sensitive secret s .

Problem

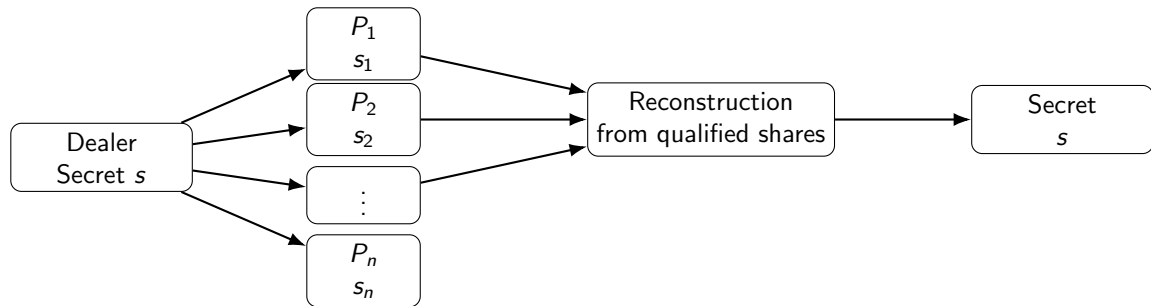
How can the dealer distribute the secret among n parties such that the secret is protected even if some parties are compromised?

We want to divide the secret into multiple **shares** such that:

- No individual party can recover the secret.
- A sufficiently large set of parties can recover the secret.
- The secret is not revealed to unauthorized parties.

Secret s $\longrightarrow$ Shares $\longrightarrow$ Secret s

Secret Sharing: Problem Description



The dealer distributes:

$$s_1, s_2, \dots, s_n$$

A qualified set of shares should reconstruct:

$$s$$

- Individual shares should reveal no information about s .
- Only an authorized set of parties should reconstruct s .

Additive Secret Sharing

Additive Secret Sharing: The Problem

Suppose a dealer has a secret s and wants to share it between two parties P_1 and P_2 .

Question

How can we create two shares such that:

- P_1 alone cannot recover s ,
- P_2 alone cannot recover s ,
- P_1 and P_2 together can recover s ?

Share s between P_1 and P_2 without revealing s

Additive Secret Sharing: Construction

Let s be the secret.

Choose a random value x_1 .

$$x_1 \xleftarrow{\$} \mathbb{F}$$

Then compute the second share as:

$$x_2 = s - x_1$$

The two shares are:

$P_1 : x_1$	$P_2 : x_2$
-------------	-------------

When both parties come together:

$$x_1 + x_2 = x_1 + (s - x_1) = s$$

$s = x_1 + x_2$

Why Does One Share Reveal Nothing?

Consider P_1 's share x_1 .

x_1 is chosen uniformly at random and independently of the secret s .

$$x_1 \xleftarrow{\$} \mathbb{F}$$

Therefore, seeing x_1 alone gives no information about s .

Similarly, P_2 sees:

$$x_2 = s - x_1$$

But since x_1 is uniformly random, x_2 is also uniformly distributed over $\mathbb{F}$.

Security

One share alone reveals no information about the secret.

1 share: no information	2 shares: recover s
-------------------------	-----------------------

Additive Secret Sharing: Generalization

Suppose a secret s is to be shared among n parties:

$$P_1, P_2, \dots, P_n$$

Choose $n - 1$ random shares:

$$x_1, x_2, \dots, x_{n-1} \xleftarrow{\$} \mathbb{F}$$

Choose the final share such that:

$$x_n = s - \sum_{i=1}^{n-1} x_i$$

Thus:

$$x_1 + x_2 + \dots + x_n = s$$

Each party P_i receives only its share x_i .

Key Idea

The shares add up to the secret.

Additive Secret Sharing: 256-bit Representation

The elements of the sharing space can be represented using **256-bit numbers**.

$$x_i, s \in \{0, 1, \dots, 2^{256} - 1\}$$

All addition and subtraction operations are performed modulo:

$$2^{256}$$

Thus, the additive sharing satisfies:

$$x_1 + x_2 + \dots + x_n \equiv s \pmod{2^{256}}$$

Example

Each share is a 256-bit value, and arithmetic is performed within the range of 256-bit numbers.

From Additive to Threshold Secret Sharing

In the previous scheme, the secret is reconstructed only when **all** n **parties** provide their shares.

$$x_1 + x_2 + \cdots + x_n = s$$

Therefore, all parties must be available for reconstruction.

Problem

In a real-world system, requiring every party to be present may be impractical.

What if **only some of the parties** are sufficient to reconstruct the secret?

Can we reconstruct s using only $t < n$ parties?

This leads to the idea of **Threshold Secret Sharing**.

Here we want:

Any t out of n parties can recover the secret.

A Very Useful Property

Suppose:

$$x = x_1 + x_2 + x_3$$

and

$$y = y_1 + y_2 + y_3$$

Each party can locally add its shares:

$$z_i = x_i + y_i$$

Then:

$$z = z_1 + z_2 + z_3 = x + y$$

Important

The parties can compute an additive sharing of $x + y$ **without reconstructing** x **or** y .

Threshold Secret Sharing

Threshold Secret Sharing

Suppose a secret is shared among n participants.
A (t, n) -**threshold secret sharing scheme** provides:

Reconstruction

Any t or more participants can reconstruct the secret.

Secrecy

Any $t - 1$ or fewer participants learn nothing about the secret.

t = threshold n = number of participants

How Can We Construct Threshold Sharing?

We need a mechanism with the following property:

**t pieces determine the secret,
but fewer than t pieces do not.**

A natural mathematical object has exactly this property:

A polynomial of degree $t - 1$.

Before moving to finite fields, let us first understand the idea using ordinary real numbers.

Polynomial Intuition

Consider a line:

$$y = a + bx$$

A line is determined by **2 points**.

Now consider a quadratic:

$$y = a + bx + cx^2$$

A quadratic is determined by **3 points**.

More generally:

Polynomial Interpolation

A polynomial of degree $t - 1$ is uniquely determined by t distinct points.

This gives us the reconstruction property we need.

Constructing Shamir's Threshold Secret Sharing

Suppose the secret is s and we want a (t, n) threshold scheme.

Choose a polynomial of degree $t - 1$:

$$f(x) = s + a_1x + \cdots + a_{t-1}x^{t-1}$$

For each party P_i :

- Choose a distinct nonzero point x_i .
- Evaluate the polynomial at x_i :

$$y_i = f(x_i)$$

- Give the share:

$$(x_i, y_i)$$

Thus:

$$P_1 \rightarrow (x_1, y_1), \quad P_2 \rightarrow (x_2, y_2), \quad \cdots, \quad P_n \rightarrow (x_n, y_n)$$

Reconstructing the Secret

Suppose t shares are available:

$$(x_1, y_1), (x_2, y_2), \dots, (x_t, y_t)$$

These t points uniquely determine the polynomial $f(x)$ of degree at most $t - 1$. Using **Lagrange interpolation**:

$$f(x) = \sum_{i=1}^t y_i \prod_{\substack{1 \leq j \leq t \\ j \neq i}} \frac{x - x_j}{x_i - x_j}$$

The secret is the value at $x = 0$:

$$s = f(0)$$

Therefore:

$$t \text{ shares} \longrightarrow \text{interpolation} \longrightarrow s$$

Security of Threshold Secret Sharing

For a (t, n) threshold scheme:

- Any t or more shares can uniquely reconstruct the secret s .
- Any $t - 1$ or fewer shares reveal no information about s .

For t shares, the polynomial of degree $t - 1$ is uniquely determined:

$$t \text{ shares} \implies \text{unique polynomial} \implies s = f(0)$$

For fewer than t shares, many degree- $t - 1$ polynomials can pass through the available points, corresponding to different possible secrets.

$$t - 1 \text{ shares} \implies \text{many possible secrets}$$

Security Property

Any $t - 1$ or fewer shares reveal no information about the secret.

Secret Sharing over Finite Fields

Cryptography needs a finite and well-defined computational domain.

Real numbers are problematic because:

- They have unbounded precision.
- Exact representation is not practical.
- Numerical approximation can introduce errors.
- Cryptographic algorithms require exact arithmetic.

Solution

Perform polynomial arithmetic over a **finite field**.

Shamir Secret Sharing: Security Conditions

For a (t, n) Shamir scheme:

1 Polynomial degree

$$\deg(f) = t - 1$$

2 Leading coefficient

$$a_{t-1} \neq 0$$

3 Random coefficients

$a_1, \dots, a_{t-2}$ are independently and uniformly random.

4 Distinct evaluation points

$x_1, \dots, x_n$ must be distinct.

5 Sufficient field size

The field must contain enough distinct evaluation points.

Information-Theoretic and Post-Quantum Security

Shamir Secret Sharing provides **information-theoretic security**.

- Any set of fewer than t shares reveals no information about the secret s .
- Security does not depend on any computational hardness assumption.
- Even an adversary with unlimited computational power cannot recover s from fewer than t shares.

Post-Quantum Security

Since the security is information-theoretic and does not rely on factoring, discrete logarithms, or other computational assumptions, Shamir Secret Sharing remains secure against quantum adversaries.

Information-Theoretic Security $\implies$ Post-Quantum Secure

Acknowledgment:

- ChatGPT (OpenAI) was used to identify the content for the slides, which was subsequently verified by me.



Dr. Laltu Sardar, Assistant Professor, IISER Thiruvananthapuram
laltu.sardar@iisertvm.ac.in, laltu.sardar.crypto@gmail.com
Course webpage: https://laltu-sardar.github.io/courses/26_crypto.html.